

Prevalence of Hypertension and Determinants of Poor Blood Pressure Control in Patients with Type 2 Diabetes Mellitus Attending a Tertiary Clinic in Lagos, Nigeria

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Abstract

Objective: The objective is to determine the prevalence of comorbid hypertension and blood pressure (BP) control among patients with Type-2-diabetes-mellitus attending a tertiary-hospital in Lagos, and identify the determinants of poor BP control. **Materials and Methods:** A cross-sectional study of 238 consecutive patients with Type-2-diabetes Mellitus (DM) at the adult diabetes-clinic of a tertiary health-facility in Lagos, Nigeria over a 5-month period. Data were retrieved with the aid of structured-investigator-administered-questionnaire, physical examination, and review of hospital record. Hypertension was defined as BP $\geq 140/90$ mmHg and target BP control was defined as $< 130/80$ mmHg. Logistic regression analysis was used to identify the independent determinants of poor BP control. **Results:** Comorbid hypertension was present in 187 (78.6%) of study participants with males (68/87 [78.8%]) and females (119/151 [78.2%]) similarly affected, $P = 0.907$. Older age (62.9 ± 10.1 vs. 54.9 ± 9.6 years) and obesity (35.3% vs. 17.6%) were associated with comorbid hypertension, $P < 0.05$. Awareness, treatment, and medication adherence rates were 96.3%, 100%, and 46%, respectively. Only 17.1% ($n = 32/187$) had BP controlled to target. Waist circumference (WC) (adjusted odd ratio: 1.04, 95% confidence interval [CI]: 1.01–1.06) and poor glycemic control (adjusted odd ratio: 5.39, 95% CI: 2.07–13.99) were the predictors of poor BP control. **Conclusion:** The prevalence of co-morbid hypertension in Type 2 DM patients in our setting is high and the BP control rate is low. Increasing WC and poor glycemic control are the independent determinants of poor BP control. Individualized weight reduction and glycemic control strategies may help achieve target BP control.

Keywords: Blood pressure control, comorbid hypertension, glycemic control, Nigeria, type 2 diabetes mellitus

Résumé

Objectif: L'objectif est de déterminer la prévalence de l'hypertension comorbide et du contrôle de la pression artérielle (PA) chez les patients atteints de diabète sucré de type 2 fréquentant un hôpital tertiaire à Lagos et d'identifier les déterminants d'un mauvais contrôle de la pression artérielle. **Matériaux et méthodes:** Une étude transversale de 238 patients consécutifs atteints de diabète sucré (DM) de type 2 à la clinique de diabète pour adultes d'un établissement de santé tertiaire à Lagos, au Nigéria, sur une période de 5 mois. Les données ont été récupérées à l'aide d'un questionnaire structuré administré par l'investigateur, d'un examen physique et d'un examen des dossiers hospitaliers. L'hypertension a été définie comme une pression artérielle $\geq 140/90$ mmHg et le contrôle de la PA cible a été défini comme une pression artérielle inférieure à 130/80 mmHg. L'analyse de régression logistique a été utilisée pour identifier les déterminants indépendants d'un mauvais contrôle de la PA. **Résultats:** L'hypertension comorbide était présente chez 187 (78,6 %) des participants à l'étude, les hommes (68/87 [78,8 %]) et les femmes (119/151 [78,2 %]) ayant été touchés de la même manière, $P = 0,907$. L'âge plus avancé ($62,9 \pm 10,1$ contre $54,9 \pm 9,6$ ans) et l'obésité (35,3 % contre 17,6 %) étaient associés à une hypertension comorbide, P était inférieur à 0,05. Les taux de sensibilisation, de traitement et d'observance thérapeutique étaient de 96,3 %, 100 %

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et 46 %, respectivement. Seulement 17,1 % ($n = 32/187$) ont vu leur tension artérielle contrôlée pour cibler. Le tour de taille (WC) (rapport impair ajusté: 1,04, intervalle de confiance à 95%[IC]: 1,01-1,06) et mauvais contrôle glycémique (rapport impair ajusté: 5,39, IC à 95%: 2,07-13,99) étaient les prédicteurs d'un mauvais contrôle de la PA. **Conclusion:** La prévalence de l'hypertension comorbide chez les patients atteints de DM de type 2 dans notre milieu est élevée et le taux de contrôle de la PA est faible. L'augmentation de la WC et un mauvais contrôle glycémique sont les déterminants indépendants d'un mauvais contrôle de la PA. Des stratégies individualisées de réduction de poids et de contrôle glycémique peuvent aider à atteindre le contrôle de la PA cible.

Mots-clés: Contrôle de la pression artérielle, hypertension comorbide, contrôle glycémique, Nigeria, diabète sucré de type 2

INTRODUCTION

Hypertension occurs commonly in patients with Type 2 diabetes mellitus (DM) and tends to accelerate the morbidity and mortality associated with DM.^[1,2] Up to 75% of the cardiovascular complications seen in patients with Type 2 DM can be attributed to comorbid hypertension.^[1,3] Tight blood pressure (BP) control in patients with Type 2 DM has been shown to markedly reduce both micro- and macrovascular complications, as well as overall morbidity and mortality.^[4] Consequently, a lower target BP level of <130–135/80 mmHg has been recommended for Type 2 DM patients with comorbid hypertension.^[5–7] The burden of comorbid hypertension in Type 2 DM however varies widely with reported prevalence rates ranging from 35% to 75% depending on the population.^[8–12] Ethnic differences; whether population or hospital-based, and the tier of health facility are some of the factors responsible for the variability in the burden/prevalence of comorbid hypertension in DM.^[13–17]

In spite of the documented benefit of tight BP control in DM and the recommendations, however, several studies from developed and developing countries alike have reported low rates of target BP control in Type 2 DM patients with rates as low as 10%–24% being reported despite generally high awareness and treatment rates.^[2,7,8] In view of the desirability of reducing overall cardiovascular risk among patients with Type 2 DM, it is pertinent to identify individuals with comorbid hypertension who are at increased risk of adverse cardiovascular outcomes as a result of poor BP control, as well as to identify the reasons for such poor BP control so that appropriate targeted approach to reverse this negative trend can be implemented.

This study aimed to document the prevalence of concomitant hypertension in individuals with Type 2 DM attending the diabetes clinic of a tertiary hospital in Lagos, determine the proportion with BP controlled to target, and identify the determinants of poor BP control in such individuals.

MATERIALS AND METHODS

Study design, location, and participants

This was a cross-sectional study of 238 Type 2 DM patients conducted at the adult diabetes out-patient clinic of a tertiary hospital in Lagos, South-West, Nigeria. The tertiary health facility receives referrals from all over Lagos and neighboring states.

Participants were consecutive consenting known Type 2 DM patients who must have been attending the clinic for a

minimum of 1 year. They must have had laboratory assessment of serum glycated hemoglobin (HbA1c) within the 3-month period preceding study recruitment.

Patients who had been diagnosed as having Type 1 DM, diabetes secondary to known medical conditions, and pregnant women were excluded from the study.

The study was conducted between September 1, 2017, and January 31, 2018.

We obtained ethics approval from the Lagos University Teaching Hospital Health Research Ethics Committee (ADM/DCST/HREC/APP/1630); and informed consent from each individual participant. The study protocol was carried out in accordance with the Declaration of Helsinki.

Data collection method

Data were obtained with the aid of a structured investigator-administered questionnaire which was in three sections. The section consisted information such as bio-data, duration of diagnosis of diabetes, medication history. Section two assessed individual's adherence to prescribed antihypertensive medication using the four structured questions in the Simplified Medication Adherence Questionnaire with dichotomous (yes/no) responses:^[18,19] (1) Do you ever forget to take your medicine? (2) Are you careless at times about taking your medicine? (3) When you feel better, do you sometimes stop taking your medicine? (4) If you sometimes feel worse when taking the medicine, do you stop taking it? Section three documented the clinical data: weight, height, waist circumference (WC), body mass index (BMI), BP and the serum glycosylated hemoglobin (HbA1c) levels. Weight, height, WC, and BP measurements were carried out according to standard protocol.^[20–22] We measured weight to the nearest 0.1 kg on a calibrated digital balance and height to the nearest 1 cm on a mobile stadiometer with the individual in minimal clothing, bare feet, and no head-wears. With the aid of a nonstretchable tape we measured WC to the nearest 0.1 cm mid-way between the lowest rib margin and the iliac crest. BMI was calculated as weight (kg) divided by the square of height in meters. BP was measured using a mercury sphygmomanometer (Accoson®, London), following a minimum of 10 min rest with the participant seated and arm supported at the level of the heart. The first and fifth Korotkoff sounds were taken as the systolic BP (SBP) and diastolic BP (DBP), respectively. Initial measurement was done on both arms to identify the arm with the higher BP reading. At least two measurements were subsequently taken from the arm with the

higher BP reading and the mean documented as the individual's BP. We retrieved the result of serum HbA1c level (done within the 3-month period preceding study recruitment) of each individual from the hospital records.

Operational definitions

- Hypertension was defined as SBP ≥ 140 mmHg and/or DBP ≥ 90 mmHg, or lower in individuals who were already on antihypertensive medication^[6]
- Target BP control was defined as BP $< 130/80$ mmHg^[7]
- Obesity was defined as BMI ≥ 30.0 ^[20]
- Poor glycemic control was defined as a serum glycated hemoglobin (HbA1c) level $\geq 7\%$ ^[23]
- Nonadherence to prescribed antihypertensive medication was defined as a "yes" response to any of the questions on the simplified medication adherence questionnaire.^[18,19]

Statistical analysis

Data obtained were analyzed using IBM Statistical software Package for Social Sciences (SPSS) version 20.0 (IBM Corp., Armonk, NY, USA). Continuous variables are presented

as mean $\pm$ standard deviation or median (interquartile range [IQR]) as appropriate while categorical variables are presented as frequencies (percentages). Comparison between means was done using the Student's *t*-test for normally distributed variable, while the median of nonnormally distributed variable was compared using the Mann-Whitney *U*-test. Comparison between categorical variables was done using the Chi-square test. Fisher's exact test was used for comparison where expected frequency was < 5 . Univariable and multivariable logistic regression analysis was used to evaluate factors that were independently associated with poor BP control. The level of statistical significance was set at a $P < 0.05$.

RESULTS

Baseline characteristics and prevalence of co-morbid hypertension

A total of 238 individuals with Type 2 DM comprising 87 (36.5%) males and 151 (63.5%) females were studied. Their age ranged from 30 to 89 years with a mean of

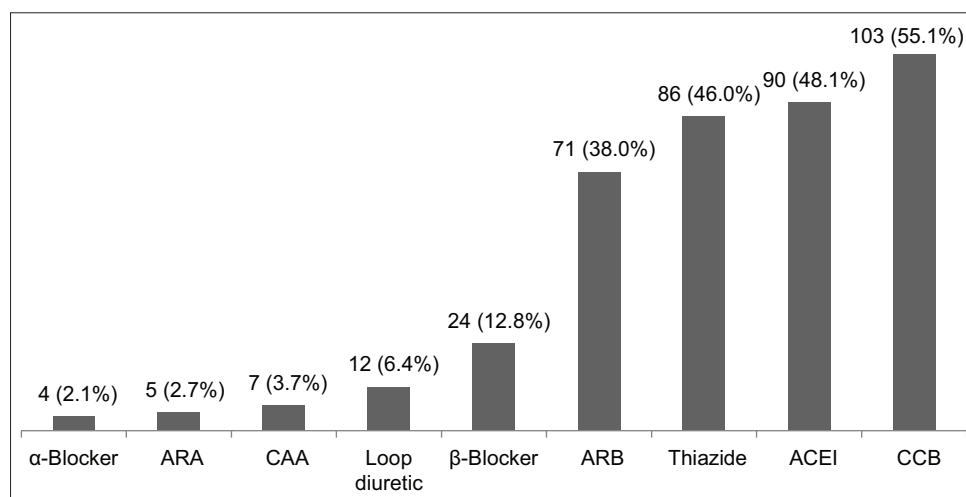


Figure 1: Classes of antihypertensive medications prescribed to patients with comorbid hypertension α -blocker = α -adrenergic receptor antagonist, β -blocker = β -adrenergic receptor antagonist, ARA = aldosterone receptor antagonists, ARB = angiotensin II receptor antagonist, ACEI = angiotensin converting enzyme inhibitor, CAA = centrally acting adrenergic antagonists, CCB = calcium channel blocker

Table 1: Comparison of participants with and without co-morbid hypertension

Characteristic	Mean $\pm$ SD		P
	Co-morbid hypertension (n=187)	No hypertension (n=51)	
Age (years)	62.9 $\pm$ 10.1	54.9 $\pm$ 9.6	<0.001
Pill burden	6.0 $\pm$ 2.5	4.6 $\pm$ 2.2	<0.001
Waist circumference (cm)	98.3 $\pm$ 20.7	93.0 $\pm$ 11.6	0.078
BMI (kg/m ²)	28.5 $\pm$ 4.8	25.6 $\pm$ 3.7	<0.001
SBP (mmHg)	134.3 $\pm$ 17.1	118.6 $\pm$ 11.8	<0.001
DBP (mmHg)	79.6 $\pm$ 11.6	74.8 $\pm$ 7.4	<0.001
Duration of diagnosis of diabetes (years), median (IQR)	10.0 (9.0)	7.0 (10.0)	0.073
HbA1c (%), median (IQR)	7.3 (1.6)	7.6 (3.1)	0.448
Obesity (BMI ≥ 30 kg/m ²), frequency (%)	66 (35.3)	9 (17.6)	0.016

HbA1c=Glycosylated haemoglobin, BMI=Body mass index, IQR=Interquartile range, SBP=Systolic blood pressure, DBP=Diastolic blood pressure, SD=Standard deviation

61.2 ± 0.7 years. The mean age of the males (61.7 ± 9.9 years) and females (60.9 ± 10.8 years) were comparable, $P = 0.555$.

Co-morbid hypertension was present in 187 (78.6%) of all the participants, with males (68/87 [78.8%]) and females (119/151 [78.2%]) being similarly affected, $P = 0.907$.

As shown in Table 1, individuals with comorbid hypertension were older, had significantly higher BMI and higher frequency of obesity.

Of the 187 individuals with comorbid hypertension, 180 (96.3%) were previously aware of their hypertension status. All (100%) previously known hypertensive individuals were prescribed 1–4 different antihypertensive medications, median: 2 (IQR: 2). Of these, 133 (73.9%) were prescribed at least 2 antihypertensive medications. Figure 1 shows the classes of prescribed antihypertensive medications. Calcium

channel blockers (CCB) (55.1%), angiotensin-converting enzyme inhibitors (48.1%), thiazide diuretics (46.0%), and angiotensin receptor antagonists (38.0%) were the most commonly prescribed. The rennin-angiotensin-aldosterone system inhibitors (RAAS) group made up 161 (86.1%) of all the prescribed antihypertensive medications. Spironolactone and α -methyldopa were the only prescribed aldosterone receptor antagonist and centrally acting adrenergic antagonist, respectively.

Overall, only 86 (46.0%) were adherent to prescribed antihypertensive medications while 101 (54.0%) were nonadherent.

Prevalence of target blood pressure control

Only 17.1% ($n = 32/187$) of individuals with comorbid hypertension had BP controlled to target. As shown in

Table 2: Comparison of participants with co-morbid hypertension who had controlled and uncontrolled blood pressures

Characteristic	Frequency		<i>P</i>
	BP controlled ($n=32$), n (%)	BP uncontrolled ($n=155$), n (%)	
Gender			
Male	10 (31.2)	58 (37.4)	0.509
Female	22 (68.8)	97 (62.6)	
Obesity (BMI ≥ 30 kg/m ²)	9 (28.1)	57 (36.8)	0.351
Number of antihypertensive medication			
0	0 (0)	7 (4.5)	0.583*
1	9 (28.1)	38 (24.5)	
2	9 (28.1)	57 (36.8)	
3	10 (31.2)	41 (26.5)	
4	4 (12.5)	12 (7.7)	
Class of antihypertensive combination for ≥ 2 drugs ($n=133$)			
RAAS + CCB based	4 (17.4)	27 (24.5)	0.147*
RAAS + diuretic based	3 (13.0)	33 (30.0)	
RAAS + CCB + diuretic based	13 (56.5)	43 (39.1)	
Other combinations	3 (13.0)	7 (6.4)	
Adherence to medication			
Yes	13 (40.6)	73 (47.1)	0.504
No	19 (59.4)	82 (52.9)	
Presence of poor glycemic control (HbA1c $\geq 7\%$)	17 (53.1)	120 (77.4)	0.005
Feel they are spending too much on medications (yes)	19 (59.4)	111 (71.6)	0.178
Feel they are taking too many medicines (yes)	12 (37.5)	68 (43.9)	0.563
Experience untoward effects from their medications	5 (15.6)	35 (22.6)	0.353
Feel the medications are effective in treating their illness	28 (87.5)	143 (92.3)	0.482
Experience difficulty accessing their medications	12 (38.4)	57 (36.8)	0.769
Age (years), mean $\pm$ SD	63.7 $\pm$ 10.0	64.7 $\pm$ 10.5	0.633
Pill burden, mean $\pm$ SD	5.8 $\pm$ 2.3	6.0 $\pm$ 2.5	0.568
Waist circumference (cm) , mean $\pm$ SD	90.1 $\pm$ 24.3	99.9 $\pm$ 22.6	0.048
BMI (kg/m ²), mean $\pm$ SD	27.8 $\pm$ 4.9	28.6 $\pm$ 6.3	0.580
SBP (mmHg), mean $\pm$ SD	118.9 $\pm$ 13.1	137.7 $\pm$ 16.1	<0.001
DBP (mmHg), mean $\pm$ SD	66.3 $\pm$ 7.8	82.5 $\pm$ 10.2	<0.001
Duration of diagnosis of diabetes (years), median (IQR)	10.0 (12.5)	9.0 (8.0)	0.135
Monthly income (Naira), median (IQR)	100,000.00 (47,500.00)	50,000.00 (40,000.00)	<0.001
Amount spent on medication monthly (Naira), median (IQR)	20,000.00 (20,000.00)	15,000.00 (11,700.00)	0.524
HbA1c (%), median (IQR)	7.0 (2.0)	7.0 (1.0)	0.021

*Fischer's exact. HbA1c=Glycosylated hemoglobin, BMI=Body mass index, RAAS=Rennin angiotensin aldosterone system inhibitors, CCB=Calcium channel blockers, IQR=Interquartile range, SBP=Systolic blood pressure, DBP=Diastolic blood pressure, SD=Standard deviation, BP=Blood pressure

Table 3: Logistic regression for independent determinants of poor blood pressure control in individuals with comorbid hypertension

Independent variable	Crude odds ratio	95% CI	P	Adjusted odds ratio	95% CI	P
Age (years)	0.990	0.953-1.029	0.617	1.016	0.972-1.063	0.480
Gender						
Female	1.000	Reference	Reference	1.000	Reference	Reference
Male	1.315	0.582-2.973	0.510	1.830	0.701-4.776	0.217
Duration of diagnosis of diabetes (years)	0.956	0.915-1.000	0.049	0.954	0.901-1.010	0.107
Level of education						
Tertiary	1.000	Reference	Reference	1.000	Reference	Reference
Below tertiary	0.850	0.391-1.850	0.682	0.686	0.274-1.717	0.420
Average monthly income	1.000	1.000-1.000	0.214	1.000	1.000-1.000	0.130
BMI	1.047	0.962-1.139	0.289	1.061	0.957-1.176	0.259
Waist circumference	1.025	1.005-1.046	0.013	1.036	1.012-1.061	0.003
Number of antihypertensive medication	0.811	0.566-1.163	0.255	0.656	0.422-1.021	0.062
Adherence to medication	1.000	Reference	Reference	1.000	Reference	Reference
Nonadherence to medication	0.769	0.355-1.664	0.504	0.873	0.364-2.091	0.760
Good glycemic control (HbA1c <7%)	1.000	Reference	Reference	1.000	Reference	Reference
Poor glycemic control (HbA1c ≥7%)	3.025	1.373-6.665	0.006	5.386	2.073-13.990	0.001

$\chi^2=29.431$, model fit $P=0.002$. Nagelkerke $R^2=0.243$. HbA1c: Glycated hemoglobin, BMI=Body mass index, CI=Confidence interval

Table 2, there was no significant difference in medication adherence rates of individuals with and without target BP control, $P = 0.502$. Participants with uncontrolled BP had significantly higher WC compared with those with controlled BP (99.91 ± 22.56 vs. 90.06 ± 24.27 , $P = 0.048$).

Determinants of poor blood pressure control

Table 3 shows the logistic regression analysis to identify the independent determinants of poor BP control in individuals with Type 2 DM and comorbid hypertension. Only increasing WC (adjusted OR: 1.04, 95% confidence interval [CI]: 1.012–1.061, $P = 0.003$) and poor glycemic control (adjusted OR: 5.39, 95% CI: 2.07–13.99, $P = 0.001$) were independently associated with poor BP control. Participants who had poor glycemic control had about five-fold odds of having poor BP control as compared to participants with good glycemic control. Furthermore, for every unit increase in WC, the odds of having poor BP control increased by 4%.

DISCUSSION

This study aimed at documenting the prevalence of hypertensive comorbidity and identifying the predictors of poor BP control among Type 2 diabetes patients attending a tertiary health facility in Lagos, Nigeria. We found a high prevalence of comorbid hypertension of about 78.6% among patients with Type 2 diabetes which is consistent with the frequent co-existence of the two conditions possibly due to shared etiological mechanisms such as lifestyle factors (weight gain, physical inactivity, and westernized dietary habits), insulin resistance and inappropriate activation of the RAAS.^[24,25] The prevalence of comorbid hypertension among our participants is however higher than the prevalence reported by other Nigerian researchers that ranged between 58.7% and 73.4%.^[13-16] It is possible that this high prevalence reflects a growing burden of

noncommunicable diseases in the general population in Nigeria as has been projected to occur as developing countries adopt western lifestyles.^[26] Two facts support this view; first, there appears to have been a progressive rise in the prevalence of comorbid hypertension among patients with diabetes in Lagos with a reported prevalence of 41.9% about 17 years ago and 67.3% 13 years later in a similar population.^[15,27] Second, Lagos is the economic capital of Nigeria with its population being cosmopolitan and having the highest per capita income in the country.^[28] A higher inclination towards a western lifestyle among its residents with the attendant health implications may therefore be expected. Of note also is that the prevalence of co-morbid hypertension in this study closely approach the average prevalence of 70%–80% reported in most Western countries but higher than the recorded prevalence of 55% in Ethiopia and 66.7% in a combined population of Type 1 and 2 diabetics in Cameroon.^[2,8,29-31]

Similar to other studies, individuals with comorbid hypertension were older.^[2,8,27] This observation may be due to the consistently reported increases with age in the prevalence of both Type 2 diabetes and hypertension in the general population.^[32,33] Individuals with comorbid hypertension also had significantly higher BMI and frequency of obesity in keeping with the documented association of obesity with diabetes and hypertension with respect to their shared mechanistic basis especially in the setting of metabolic syndrome.^[34-36] The relatively high proportion of obesity in individuals with comorbid hypertension in this study is therefore of concern as the co-existence of obesity, hypertension, and DM has been associated with increased risk of both cardiovascular and all-cause mortality.^[37]

The prevalence (17.1%) of target BP control in individuals with comorbid hypertension in our study population is quite

low in spite of the high awareness (96.3%) and treatment rates (100%). This pattern is similar to that reported in several other studies, notwithstanding that some of the studies even employed a higher target for BP control.^[2,3,13,34,38] The low BP control rate we observed was in spite of 74% of treated individuals being prescribed combination therapy of at least two antihypertensive agents comprising the RAAS blockers, CCBs, and thiazide diuretics; which are in accordance with recommended best practices.^[6,39,40] Possible explanation for this observation may be the fact that factors other than antihypertensive drug prescription are known to also modulate BP control.^[6,39] For example, adherence to the prescribed medication is fundamental if such medication is to effectively control BP.^[6,41] Therefore, similar to previous studies, the overall poor medication adherence rate in our study may have contributed to the low BP control rate.^[42,43] However, the lack of difference in medication adherence rates of comorbid hypertensives with and without target BP control suggests the possibility of other contributory factors.

In this study, increasing WC and poor glycemic control were the independent determinants of poor BP control; which compares with reports from studies in other parts of the world.^[34,39,43-46] It is also consistent with the common co-occurrence of poor-glycemic and BP controls in Type 2 DM individuals observed across tertiary health facilities in Nigeria.^[47] This association could be explained by the fact that both increasing WC and poor glycemia are direct stimulants of elevated BP through the ability to cause vascular remodeling among others.^[48,49] Furthermore, both increasing WC and poor glycemia, as well as elevated BP are strongly linked with lifestyle habits and are all favorably amenable to healthy lifestyle modifications.^[39,50-53] The association of increasing WC and poor glycemic control with sub-optimal BP in this study may therefore suggest the presence of common practice of unfavorable lifestyle habits in this population. This view is supported by the widely reported low level of knowledge and practice of lifestyle modifications among hypertensive and diabetic individuals in Nigeria.^[42,54,55]

Although both male and female gender, lower educational status, older age, unemployed status, shorter duration of hypertension, and higher pill burden, have variously been associated with poor BP control in individuals with comorbid hypertension and diabetes, we found no such associations in our study population.^[3,4,43,46,56]

We acknowledge some study limitations: while majority of the laboratory investigations were performed at the central laboratory of the hospital, a few were carried out at laboratories outside the hospital and raise issues of inter-laboratory variability. The outside laboratories were however also those certified by the International Standard Organization. We did not obtain data on the lifestyle practices of our study population and understand that this gap in knowledge has not been addressed in this study.

CONCLUSION

There is a high prevalence of co-morbid hypertension among patients with Type 2 diabetes in our practice and the target BP control rate is low despite high awareness and treatment rates. Increasing WC and poor glycemic control are the independent determinants of poor BP control. Individualized strategies that promote healthy waist circumference and stringent glycemic control may help in achieving target BP control.

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Conflicts of interest

There are no conflicts of interest.

REFERENCES

1. Sowers JR, Epstein M, Frohlich ED. Diabetes, hypertension, and cardiovascular disease: An update. *Hypertension* 2001;37:1053-9.
2. Choukem SP, Kengne AP, Dehayem YM, Simo NL, Mbanya JC. Hypertension in people with diabetes in sub-Saharan Africa: Revealing the hidden face of the iceberg. *Diabetes Res Clin Pract* 2007;77:293-9.
3. Adler AI, Stratton IM, Neil HA, Yudkin JS, Matthews DR, Cull CA, *et al.* Association of systolic blood pressure with macrovascular and microvascular complications of type 2 diabetes (UKPDS 36): Prospective observational study. *BMJ* 2000;321:412-9.
4. Tight blood pressure control and risk of macrovascular and microvascular complications in type 2 diabetes: UKPDS 38. UK Prospective Diabetes Study Group. *BMJ* 1998;317:703-13.
5. Vijan S, Hayward RA. Treatment of hypertension in type 2 diabetes mellitus: Blood pressure goals, choice of agents, and setting priorities in diabetes care. *Ann Intern Med* 2003;138:593-602.
6. Chobanian AV, Bakris GL, Black HR, Cushman WC, Green LA, Izzo JL Jr., *et al.* The seventh report of the joint National Committee on Prevention, Detection, Evaluation, and Treatment of high blood pressure: The JNC 7 report. *JAMA* 2003;289:2560-72.
7. Bangalore S, Kumar S, Lobach I, Messerli FH. Blood pressure targets in subjects with type 2 diabetes mellitus/impaired fasting glucose: Observations from traditional and Bayesian random-effects meta-analyses of randomized trials. *Circulation* 2011;123:2799-810.
8. Geiss LS, Rolka DB, Engelgau MM. Elevated blood pressure among U.S. adults with diabetes, 1988-1994. *Am J Prev Med* 2002;22:42-8.
9. Turner R, Holman R, Matthews D, Bassett P, Coster R, Stratton I, *et al.* Hypertension in Diabetes Study (HDS): I. Prevalence of hypertension in newly presenting type 2 diabetic patients and the association with risk factors for cardiovascular and diabetic complications. *J Hypertens* 1993;11:309-17.
10. Kabakov E, Norymberg C, Osher E, Koffler M, Tordjman K, Greenman Y, *et al.* Prevalence of hypertension in type 2 diabetes mellitus: Impact of the tightening definition of high blood pressure and association with confounding risk factors. *J Cardimetab Syndr* 2006;1:95-101.
11. Maheswary N, Naveed S, Ali Z, Ahmed SM, Yousuf AM. Prevalence of hypertension in type 2 diabetics. *Pak J Physiol* 2016;12:31-4.
12. Mengesha AY. Hypertension and related risk factors in type 2 diabetes mellitus (DM) patients in Gaborone City Council (GCC) clinics, Gaborone, Botswana. *Afr Health Sci* 2007;7:244-5.
13. Unadike BC, Eregie A, Ohwovoriole AE. Prevalence of hypertension amongst persons with diabetes mellitus in Benin City, Nigeria. *Niger J Clin Pract* 2011;14:300-2.
14. Anakwue R, Arodiwe E, Ofoegbu E. The prevalence and control of hypertension among patients with type 2 diabetes mellitus in Nigeria. *J Coll Med* 2013;17:11-24.
15. Bello BT, Amira CO. Pattern and predictors of urine protein excretion among patients with type 2 diabetes attending a single tertiary hospital in Lagos, Nigeria. *Saudi J Kidney Dis Transpl* 2017;28:1381-8.
16. Aniedi U, Ekong A, Idung A. Hypertension among type 2 diabetic

- patients in Uyo, South East Nigeria. *J Adv Med Med Res* 2018;26:1-6.
17. Blaychfeld-Magnazi M, Knobler H, Voet H, Reshef N, Weitzman S, Sumner AE, *et al.* Ethnic variation in the association of hypertension with type 2 diabetes. *J Clin Hypertens (Greenwich)* 2017;19:184-9.
 18. Knobel H, Alonso J, Casado JL, Collazos J, González J, Ruiz I, *et al.* Validation of a simplified medication adherence questionnaire in a large cohort of HIV-infected patients: The GEEMA Study. *AIDS* 2002;16:605-13.
 19. Taslim BB, Oluwafemi OS. Self-reported medication adherence and blood pressure control rates in patients with chronic kidney disease. *Afr J Med Health Sci* 2015;14:61-5.
 20. Garrow JS, Webster J. Quetelet's index (W/H²) as a measure of fatness. *Int J Obes* 1985;9:147-53.
 21. Beevers G, Lip GY, O'Brien E. ABC of hypertension: Blood pressure measurement. Part II-conventional sphygmomanometry: Technique of auscultatory blood pressure measurement. *BMJ* 2001;322:1043-7.
 22. World Health Organisation. Waist Circumference and Waist-Hip Ratio. Report of a WHO Expert Consultation. Geneva, 8-11 December, 2008: WHO; 2008. Available from: <http://Waist.circumference.and.waist-hip.ratio:report.of.a.WHO.expert.consultation>. [Last accessed on 2021 Apr 07].
 23. American Diabetes Association. 11. Older adults: Standards of medical care in diabetes-2018. *Diabetes Care* 2018;41:S119-25.
 24. Long AN, Dagogo-Jack S. Comorbidities of diabetes and hypertension: Mechanisms and approach to target organ protection. *J Clin Hypertens (Greenwich)* 2011;13:244-51.
 25. Oktay AA, Akturk HK, Jahangir E. Diabetes mellitus and hypertension: A dual threat. *Curr Opin Cardiol* 2016;31:402-9.
 26. Global Status Report on Non-Communicable Diseases 2010: Description of the Global Burden of NCDs, Their Risk Factors and Determinants. WHO; 2011. Available from: https://www.who.int/nmh/publications/ncd_report_full_en.pdf?ua=1. [Last accessed on 2021 Apr 07].
 27. Okubadejo NU, Fasanmade OA. Concomitant hypertension and type 2 diabetes mellitus in Nigerians: Prevalence of obesity and its indices compared to normotensive diabetics. *Niger Med J* 2004;45:48-59.
 28. Top 20 Richest States in Nigeria in 2019. Available from: https://www.legit.ng/ask_legit. [Last accessed on 2019 Jun 15].
 29. Colosia AD, Palencia R, Khan S. Prevalence of hypertension and obesity in patients with type 2 diabetes mellitus in observational studies: A systematic literature review. *Diabetes Metab Syndr Obes* 2013;6:327-38.
 30. Tadesse K, Amare H, Hailemariam T, Gebremariam T. Prevalence of hypertension among patients with type 2 diabetes mellitus and its socio demographic factors in Nigist Ellen Mohamed Memorial Hospital Hosanna, Southern Ethiopia. *J Diabetes Metab* 2018;9:4-10.
 31. Iglay K, Hannachi H, Joseph Howie P, Xu J, Li X, Engel SS, *et al.* Prevalence and co-prevalence of comorbidities among patients with type 2 diabetes mellitus. *Curr Med Res Opin* 2016;32:1243-52.
 32. Ekpenyong CE, Akpan UP, Ibu JO, Nyebruk DE. Gender and age specific prevalence and associated risk factors of type 2 diabetes mellitus in Uyo metropolis, South Eastern Nigeria. *Diabetol Croat* 2012;41:17-28.
 33. Ogah OS, Okpechi I, Chukwuonye II, Akinyemi JO, Onwubere BJ, Falase AO, *et al.* Blood pressure, prevalence of hypertension and hypertension related complications in Nigerian Africans: A review. *World J Cardiol* 2012;4:327-40.
 34. Abougambou SS, Abougambou AS. A study evaluating prevalence of hypertension and risk factors affecting on blood pressure control among type 2 diabetes patients attending teaching hospital in Malaysia. *Diabetes Metab Syndr* 2013;7:83-6.
 35. Akalu Y, Belsti Y. Hypertension and its associated factors among type 2 diabetes mellitus patients at Debre Tabor General Hospital, Northwest Ethiopia. *Diabetes Metab Syndr Obes* 2020;13:1621-31.
 36. Hirani V, Zaninotto P, Primates P. Generalised and abdominal obesity and risk of diabetes, hypertension and hypertension-diabetes co-morbidity in England. *Public Health Nutr* 2008;11:521-7.
 37. Mottillo S, Filion KB, Genest J, Joseph L, Pilote L, Poirier P, *et al.* The metabolic syndrome and cardiovascular risk a systematic review and meta-analysis. *J Am Coll Cardiol* 2010;56:1113-32.
 38. Khunti K, Cieriello A, Cos X, De Block C. Achievement of guideline targets for blood pressure, lipid, and glycaemic control in type 2 diabetes: A meta-analysis. *Diabetes Res Clin Pract* 2018;137:137-48.
 39. Flack JM, Sica DA, Bakris G, Brown AL, Ferdinand KC, Grimm RH Jr., *et al.* Management of high blood pressure in Blacks: An update of the International Society on Hypertension in Blacks consensus statement. *Hypertension* 2010;56:780-800.
 40. Arauz-Pacheco C, Parrott MA, Raskin P, American Diabetes Association. Treatment of hypertension in adults with diabetes. *Diabetes Care* 2003;26 Suppl 1:S80-2.
 41. Piercefield EW, Howard ME, Robinson MH, Kirk CE, Ragan AP, Reese SD. Antihypertensive medication adherence and blood pressure control among central Alabama veterans. *J Clin Hypertens (Greenwich)* 2017;19:543-9.
 42. Okwuonu CG, Ojima NE, Okaka EI, Akemokwe FM. Patient-related barriers to hypertension control in a Nigerian population. *Int J Gen Med* 2014;7:345-53.
 43. Muleta S, Melaku T, Chelkeba L, Assefa D. Blood pressure control and its determinants among diabetes mellitus co-morbid hypertensive patients at Jimma University medical center, South West Ethiopia. *Clin Hypertens* 2017;23:29.
 44. Feldstein AC, Nichols GA, Smith DH, Stevens VJ, Bachman K, Rosales AG, *et al.* Weight change in diabetes and glycemic and blood pressure control. *Diabetes Care* 2008;31:1960-5.
 45. Al-Asadi JN, Al-Asady OG. Hypertension and blood pressure pattern among type 2 diabetic patients. *Thi Qar Med J* 2015;9:14-24.
 46. Dedefo MG, Gemechu DB, Fekadu G, Tekle Dibessa T. Blood pressure control among hypertensive diabetic patients on follow-up at chronic clinic of Nekemte referral hospital in West Ethiopia. *Int J Hypertens* 2020;2020:7526257.
 47. Chinenye S, Uloko AE, Ogbera AO, Ofoegbu EN, Fasanmade OA, Fasanmade AA, *et al.* Profile of Nigerians with diabetes mellitus – Diabcare Nigeria Study Group (2008): Results of a multicenter study. *Indian J Endocrinol Metab* 2012;16:558-64.
 48. Brands MW, Hopkins TE. Poor glycemic control induces hypertension in diabetes mellitus. *Hypertension* 1996;27:735-9.
 49. Ohishi M. Hypertension with diabetes mellitus: Physiology and pathology. *Hypertens Res* 2018;41:389-93.
 50. May AM, Romaguera D, Travier N, Ekelund U, Bergmann MM, Kaaks R, *et al.* Combined impact of lifestyle factors on prospective change in body weight and waist circumference in participants of the EPIC-PANACEA study. *PLoS One* 2012;7:e50712.
 51. Bralić Lang V, Bergman Marković B, Vrdoljak D. The association of lifestyle and stress with poor glycemic control in patients with diabetes mellitus type 2: A Croatian nationwide primary care cross-sectional study. *Croat Med J* 2015;56:357-65.
 52. García-Molina L, Lewis-Mikhael AM, Riquelme-Gallego B, Cano-Ibáñez N, Oliveras-López MJ, Bueno-Cavanillas A. Improving type 2 diabetes mellitus glycaemic control through lifestyle modification implementing diet intervention: A systematic review and meta-analysis. *Eur J Nutr* 2020;59:1313-28.
 53. Godlee F. Pills are not the answer to unhealthy lifestyles. *BMJ* 2018;362:k3046.
 54. Okwuonu CG, Emmanuel CI, Ojima NE. Perception and practice of lifestyle modification in the management of hypertension among hypertensives in south-east Nigeria. *Int J Med Biomed Res* 2014;3:121-31.
 55. Ogundele SO, Dada AO, Mosuro OR. Clinical profile, knowledge, and beliefs about diabetes among patients attending a Tertiary Health Centre in Lagos: A cross-sectional survey. *Niger J Clin Pract* 2016;19:508-12.
 56. Adeniyi OV, Yogeswaran P, Longo-Mbenza B, Ter Goon D. Uncontrolled hypertension and its determinants in patients with concomitant type 2 diabetes mellitus (T2DM) in rural South Africa. *PLoS One* 2016;11:e0150033.